COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 10 : SLO 2 — Tutorial: Non-Restoring Division
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Problem 2: Non-Restoring Division of 0101 (5) ÷ 0010 (2)

• Init: A = 0000, Q = 0101, M = 0010

• Step 1: shift (A,Q) → A = 0000, Q = 1010; A ≥ 0 → A = A - M = 1110 (< 0), Q0 = 0 → A = 1110, Q = 1010

• Step 2: shift → A = 1101, Q = 0100; A < 0 → A = A + M = 1101 + 0010 = 1111 (< 0), Q0 = 0 → A = 1111, Q = 0100

• Step 3: shift → A = 1110, Q = 1000; A < 0 → A = A + M = 1110 + 0010 = 0000 (≥ 0), Q0 = 1 → A = 0000, Q = 1001

• Step 4: shift → A = 0001, Q = 0010; A ≥ 0 → A = A - M = 0001 - 0010 = 1111 (< 0), Q0 = 0 → A = 1111, Q = 0010

• Final correction: A < 0 → A = A + M = 1111 + 0010 = 0001

• Final Quotient = Q = 0010 (binary) = 2 (decimal)

• Final Remainder = A = 0001 (binary) = 1 (decimal)

• Verification: 5 ÷ 2 = 2 remainder 1 ✓